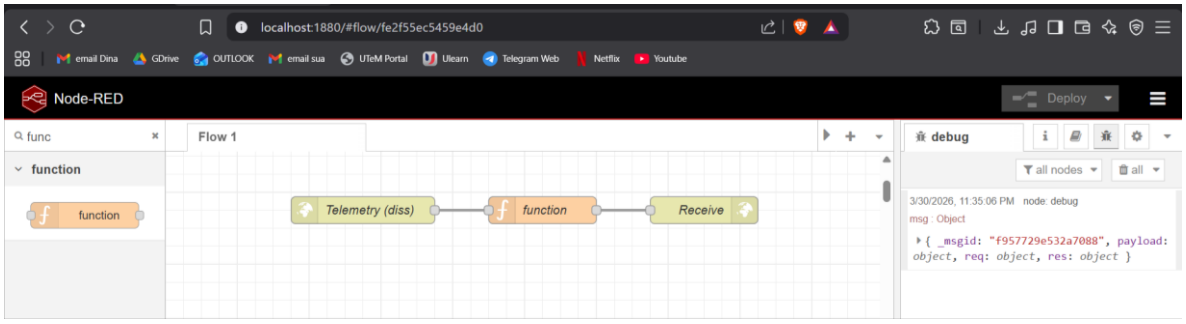
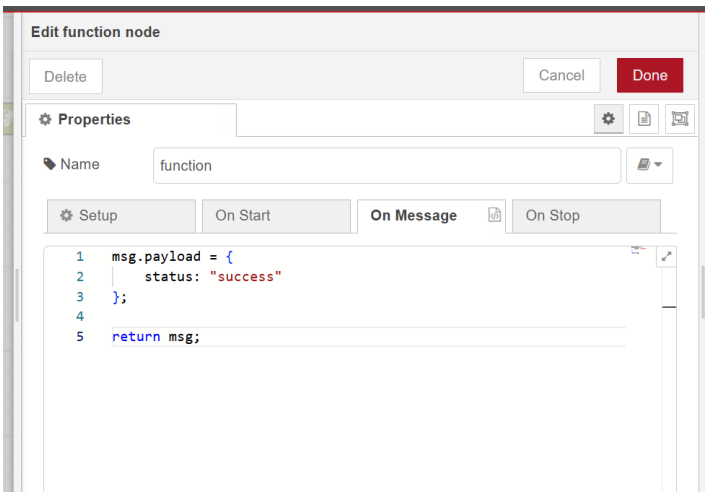
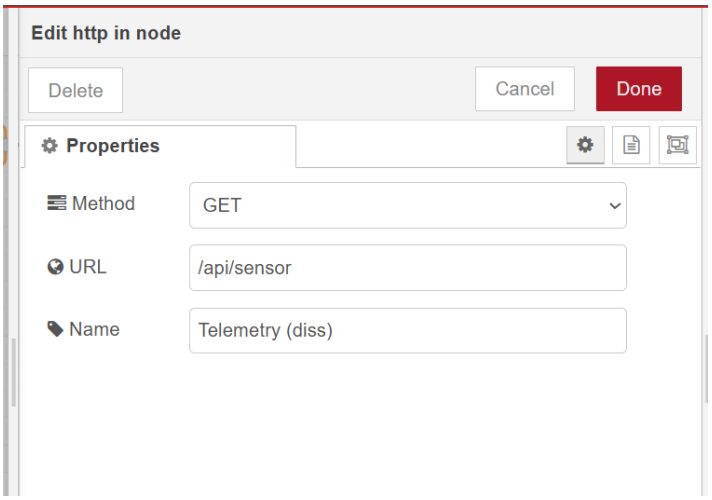
Exercise 2: Protocol Exploration & Local REST APIs

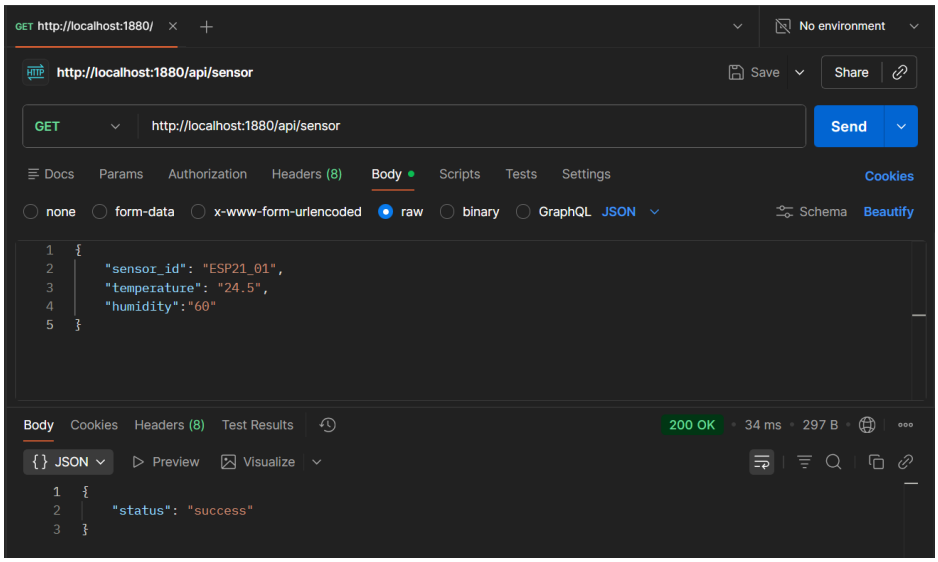
Discussion:

A: HTTP Method









Q: What happened to your JSON payload when you switched to a GET request? Did Node-RED successfully log the temperature data in the debug window? Explain the fundamental architectural difference between how a POST request and a GET request handle data payloads.

A: When the HTTP method was changed from POST to GET, the JSON payload sent from Postman was not processed by Node-RED as expected. Although a response with status 200 OK was still received, the sensor data was not logged in the debug window.

This occurred because GET requests do no support sending data in the request body whereas POST request are specifically designed to carry data in the body using formats such as JSON. In this discussion, the JSON payload containing the sensor information was entered in the body section of Postman. However, when the method was switched to GET, this payload was ignored by Node-RED.

The response observed in Postman was generated by the function node, which manually returned a predefined JSON response. Therefore, the response did not reflect any actual data received from the client, but rather a static output configured in the flow.

In conclusion, this demonstrated that POST i

<u>Fundamental Architectural Difference between POST request & GET request</u>

POST Request:

1. Sends data in the request body

2. Used for creating or sending new data
3. Supports structured payloads like JSON

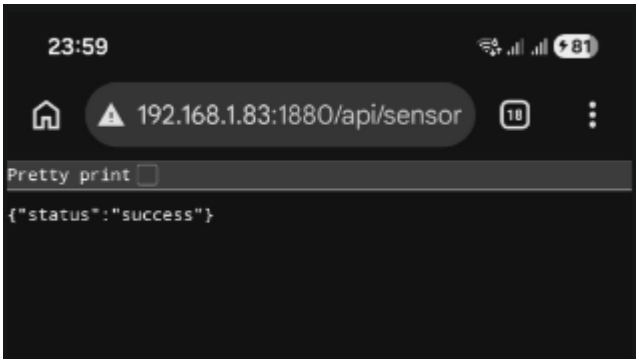
GET Request:

1. Sends data via URL parameters, not in the body
2. Used for retrieving data
3. Does not support body payload

B: Local Area Network (LAN) Edge Connectivitiy

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:d08:c2:54:b884:9c21:c9d4:f924
   Temporary IPv6 Address. . . . . . : 2001:d08:c2:54:64e4:9bea:530e:9950
   Link-local IPv6 Address . . . . . : fe80::628a:6a08:5cf4:85a%8
   IPv4 Address. . . . . . . . . . . : 192.168.1.83
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%8
                                       192.168.1.254
```



Q: Did your phone successfully trigger the Node-RED flow? Explain why you are not able to send a request to the POST method from your phone?

A: The smartphone successfully accessed the Node-RED server using the local IPv4 address. The browser displayed the response " "status": "success" " indicating that the request from the mobile device was able to read the Node-RED flow running the local machine.

This confirms that the Docker container, which is bound to port 1880 of the host machine, is functioning as a local edge server within the same Local Area Network (LAN). Since both the computer and smartphone are connected to the same Wi-Fi network, the smartphone can communicate directly with the Node-RED service using the host's IP address.

However, it is not possible to send a POST request directly from the smartphone browser. This is because web browsers typically send GET requests by default when accessing a URL. Unlike tools such as Postman, a browser does not provide a built-in interface to construct and send HTTP POST requests with a JSON payload.

As a result, while the smartphone can successfully trigger the Node-RED flow using a GET request, it cannot send structured data such as sensor telemetry through a POST request without using a specialized tool or application.

C: Deep Packet Inspection via Debugging

Analyzing both the debug message in NodeRed for both activities A and B above, expand the entire msg object by clicking the small arrows next to the logged message.

● Look deeply at the expanded msg.req (request) object. Identify and list the following components involved in your HTTP protocol transmission:

a) What is the content-type listed under the headers?

Activities in A	Activities in B
application/json ▾headers: *object* content-type: "application/json"	No content-type (no body sent)

b) What is the IP address of the client that sent the request (look for remoteAddress)?

Activities in A	Activities in B
Host: "localhost:1880" Socket: "[internal]" socket: "[internal]"	host: "192.168.1.83:1880" socket: "[internal]" client: "[internal]" socket: "[internal]" client: "[internal]"

c) Compared the differences between A and B

The content-type header specifies the format of data being transmitted from the client to the server. For the POST request, the Content-Type is identified as: application/json. This indicates that the POST request carries a message body formatted in JSON. This is expected as POST is commonly used to send structured data such as sensor readings or user inputs to the server.

In contrast, the GET request does not include a Content-Type header. This is because GET request do not contain a message body and are primarily used for retrieving data from the server rather than sending it. This difference highlights that POST requests are designed for data transmission, whereas GET request are intended for data retrieval without payload content.

For Client IP Address, it indicates the origin of the HTTP request and can be observed through the request properties such as host, client or socket. For the POST request, the host is shown as **localhost:1880** and the connection is labelled as **socket: [internal].** This indicates that the request originates from the same machine where Node-RED is running. In this case, the request was sent using Postman locally, and therefore the client IP corresponds to the loopback address.

For the GET request, the host is **192.168.1.83:1880** which represent a local network (LAN) IP address. This suggests that the request was made from an external device, such as a smartphone connected to the same network. Although the debug still shows **socket: [internal],** this reflects Node-RED's internal handling of the request rather than the true origin of the client.

This comparison shows that POST requests in this experiment originate from a local internal source, while GET requests originate from an external device within the network.

Component	POST Request	GET Request
Content-Type	application/json	Not present
Data Transmission	Sends JSON data	No body (retrieval only)
Client Source	Internal (localhost)	External device (LAN)